

Article

Seasonal Water Use Patterns of *Eucalyptus* with Different Ages in Southern Subtropical China

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Abstract: Seasonal droughts induced by climate change pose a significant threat to the normal growth patterns of forests in the subtropical regions of southern China. Therefore, it is crucial to explore the response of tree water use patterns to seasonal drought to maintain tree physiological activities. However, it remains unknown whether changes in dry and wet seasons have an impact on the water use patterns of trees of different ages. In this study, a two-year experiment was conducted in *Eucalyptus urophylla* × *Eucalyptus grandis* (hereinafter referred to as *Eucalyptus*) plantations at three ages (4, 7, and 17 years). Specifically, the water use patterns of *Eucalyptus* in dry and wet seasons were calculated using hydrogen stable isotopes (including the isotopes in xylem water and 0–150 cm soil layers) coupled with MixSIAR. The results showed that there were notable variations in the proportions of water absorption from different soil layers by *Eucalyptus* during dry and wet seasons. During the dry season (April 2024), 4-year-old and 7-year-old *Eucalyptus* primarily utilized water from the 40–90 cm soil layer, while 17-year-old *Eucalyptus* mainly relied on deep soil water at depths of 60–150 cm, with a utilization ratio of 50.9%. During the wet season (August 2023), the depth of water uptake by *Eucalyptus* of different ages significantly shifted towards shallow layers, and the trees primarily utilized surface soil water from the 0–60 cm layer, with utilization ratios of 59.9%, 64.8%, and 61.6% for 4-year-old, 7-year-old, and 17-year-old *Eucalyptus*, respectively. The water sources of *Eucalyptus* during dry and wet seasons were variable, which allowed *Eucalyptus* to cope with seasonal drought stress. The differences in the water uptake strategies of *Eucalyptus* between dry and wet seasons can be attributed to their long-term adaptation to the environment. Our research revealed the differences in the water utilization of *Eucalyptus* with various ages between dry and wet seasons in subtropical China, providing new insights for a better understanding of the adaptive mechanisms of subtropical forests in response to alterations in water conditions caused by climate change.



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1. Introduction

The water use pattern of trees is not only a key factor determining their survival, growth, and distribution, but also the foundation for forest ecosystems to fulfil functions such as material supply, water conservation, and climate regulation [1–3]. Especially in the context of frequent seasonal drought events caused by climate change, the flexible water use pattern of trees is crucial for maintaining forest biodiversity and ecosystem stability [4,5]. Forest age, a major characteristic indicator of forest stands, has a direct impact on the

water use pattern of trees in forests, thereby reflecting the adaptability of forest ecosystems to changing environments [6,7]. Therefore, exploring the water use patterns of trees of different ages is of great significance for revealing the response mechanism of forests to climate change and promoting sustainable forest development.

Despite the importance of this topic, two uncertainties still exist in studies examining the water use patterns of trees of different ages. First, the water use patterns of trees of different ages remain controversial. Generally, compared with young trees, mature trees (*Populus tomentosa* Carrière, *Pinus ponderosa* Douglas ex C. Lawson, and *Hevea brasiliensis* (Willd. ex A. Juss.) Müll. Arg.) mainly absorb water from deeper soil layers due to their deeper and wider root distribution [8–10]. However, Wang et al. [11] found that the main water absorption depth of *Malus pumila* Mill. became shallower with increasing forest age (10, 15, and 22 years); moreover, their water absorption patterns were not correlated with root distribution [11]. The above inconsistent results indicate that there is significant difference in the water use patterns of trees of different ages among diverse tree species. Therefore, it is necessary to explore the water use patterns of various tree species at different ages.

Second, it is still unclear whether seasons affect the water use patterns of trees of different ages. In general, trees primarily absorb water from shallow soil layers during the wet season, whereas they tend to utilize more water from deep soil layers during the dry season [12,13]. However, recent research has obtained opinions that are inconsistent with the aforementioned studies. For example, young *Pinus sylvestris* L. (4 years old) mainly absorbed shallow soil water during both dry and wet seasons, while older *P. sylvestris* (10 and 18 years old) exhibited a flexible water use pattern, specifically absorbing shallow and middle soil water during the wet season but middle and deep soil water during the dry season [14]. Similarly, 18-year-old *Robinia pseudoacacia* L. mainly utilized deep soil water during the dry season, while it switched to absorb shallow soil water during the wet season. However, there was no difference in the water use pattern of 30-year-old *R. pseudoacacia* between the dry and wet seasons [15]. These incompatible results imply that the water use patterns of trees of different ages vary significantly on a seasonal scale. Therefore, it is urgent to investigate the water use patterns of trees of different ages based on seasonal scales to enrich our understanding of this field.

In order to address the above issues, this study chose *Eucalyptus* plantations of different ages as objects to explore their water use patterns during the dry and wet seasons. *Eucalyptus*, an important timber tree species, is widely cultivated in tropical and subtropical regions around the world due to its fast-growing and high-yielding characteristics. The preserved area of *Eucalyptus* plantations is about 5,694,000 ha in southern China [16]. Against the backdrop of climate change, the increasing severity of seasonal drought severely limits *Eucalyptus* growth and poses a significant risk to the productivity of *Eucalyptus* plantations [17–19]. Therefore, in this study, we employed hydrogen stable isotopes coupled with the Bayesian mixture model to reveal the seasonal variations in water sources and the water use patterns of *Eucalyptus* of different ages in the low mountainous hilly areas of southern China. This study aims to answer the two following questions: (1) Does forest age affect the water use patterns of *Eucalyptus*? (2) Are there differences in the water use patterns of *Eucalyptus* in various seasons?

2. Materials and Methods

2.1. Study Area

This study was carried out at the Nanning Eucalypt Plantation Ecosystem Observation and Research Station of Guangxi, China (107°59′ E–108°18′ E, 22°28′ N–22°46′ N, Figure 1), which experiences a South Asian monsoon climate. The mean annual air temperature fluctuates between 21 °C and 22 °C. The mean annual precipitation ranges from 1200 mm

to 1300 mm, with more than 80% of the rainfall occurring from May to September. The area receives an average of 1800 h of sunshine per year. The study area is characterized primarily by hilly terrain, and the soil type is mainly lithosols. The average thickness of the soil layer is approximately 0.8 m, and the soil pH value is slightly acidic. The soil fertility is moderate.

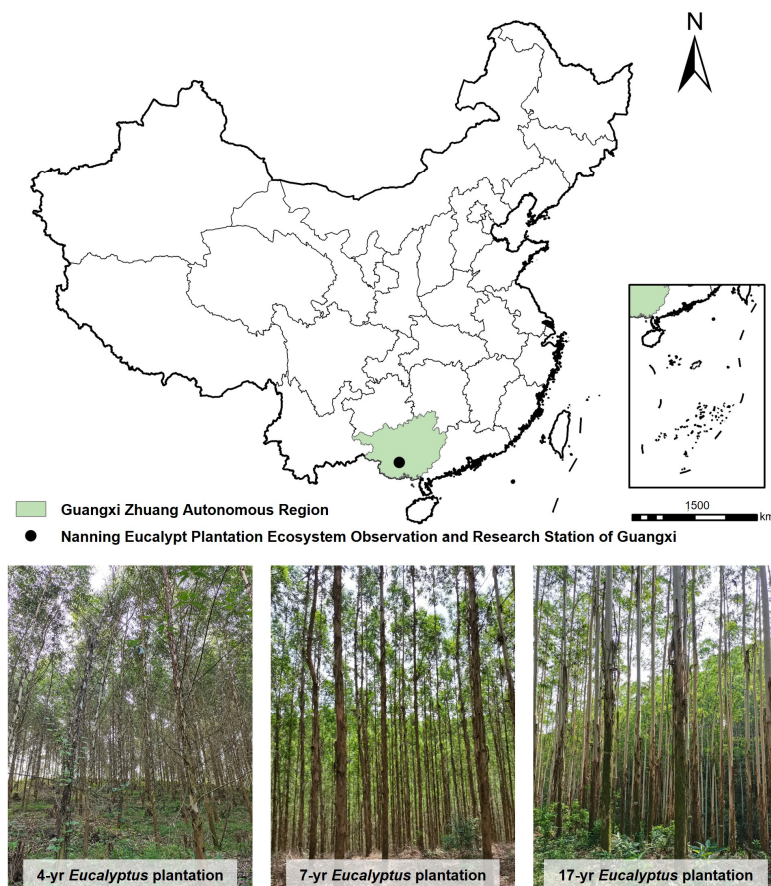


Figure 1. Location of study area and sceneries of sample sites.

The vegetation type in the experimental area is classified as subtropical evergreen broad-leaved forest, with the main forest stand consisting of *Eucalyptus* plantations. The dominant species in the arbor layer of the *Eucalyptus* plantation is *E. urophylla* × *E. grandis* (hereinafter referred to as *Eucalyptus*). The shrub layer is primarily composed of *Melastoma candidum* D. Don, *Rhodomyrtus tomentosa* (Ait.) Hassk., *Cyrtococcum patens* (L.) A. Camus, *Trema tomentosa* (Roxb.) Hara, *Rhus chinensis* Mill., and *Maesa japonica* (Thunb.) Moritzi. The herbaceous layer mainly consists of *Miscanthus sinensis* Andersson, *Hedyotis vestita* R. Br. ex G. Don, and *Commelina diffusa* Burm.f. The experiment was conducted between August 2023 and April 2024 in *Eucalyptus* plantations aged 4, 7, and 17 years. Three random plots with areas of 20 m × 20 m were selected within each *Eucalyptus* stand, ensuring a relatively uniform spatial distribution of trees. The stand characteristics of each experimental site, such as geographical location, tree height, diameter at breast height, and stand density, were investigated (Table 1). In addition, the soil texture across 0–150 cm soil layers of *Eucalyptus* plantations of different ages was also examined (Table 2). Given that the three experimental sites were all situated on hilly areas far from the groundwater level (4 m below the surface, at an elevation of 140 m), groundwater was not considered a source of plant water.

Table 1. Site characteristics of *Eucalyptus* stands of different ages.

Stand Age (a)	Geographic Location	Altitude (m)	Average DBH (cm)	Average Height (m)	Density (Tree·ha ^{−1})
4	108°11'46'' E 22°43'58'' N	176–187	12	19	1248
7	108°11'54'' E 22°41'24'' N	201–208	17	24	1112
17	108°11'43'' E 22°44'27'' N	174–181	34	37	735

Note: DBH stands for diameter at breast height.

Table 2. Soil texture at 0–150 cm depth in *Eucalyptus* plantations of different ages.

Stand Age (a)	Soil Texture (%)	Soil Depth (cm)				
		0–20	20–40	40–60	60–90	90–150
4	Clay	23.7	34.5	32.6	24.4	26.5
	Silt	46.8	35.4	26.1	38.9	34.3
	Sand	29.5	30.1	41.3	36.7	39.2
7	Clay	26.2	32.1	24.9	22.8	25.4
	Silt	38.5	37.2	40.5	36.3	30.5
	Sand	35.3	30.7	34.6	40.9	44.1
17	Clay	24.8	35.2	30.1	26.9	20.2
	Silt	33.9	40.2	36.5	38.4	47.5
	Sand	41.3	24.6	33.4	34.7	32.3

2.2. Sample Collection

2.2.1. Collection of Rainfall Samples

Precipitation in the region occurs in the form of rainfall, and we utilized rain gauges to collect rainfall samples. During the collection process, we aimed to prevent the exchange of collected rainfall with the surrounding air to minimize evaporation. Before sampling, precipitation was recorded for each rainfall event in which the precipitation exceeded 5 mm during the observation period. After each rainfall event, a rainfall sample was collected in a sampling bottle. The bottle was subsequently sealed with Parafilm and stored at 4 °C.

2.2.2. Collection of Tree Stem (Xylem) Samples

The distinction between dry and wet seasons in Nanning was clear: the wet and dry seasons occurred from May to October each year and November to April of the following year, respectively. To ensure the representativeness of the collected samples from the dry and wet seasons, wet season sampling was conducted 2 days after moderate rain (10–25 mm), and dry season sampling was conducted after 15 days with no precipitation. In each sample plot of each site, three *Eucalyptus* trees with a similar crown width and ground diameter were selected for sampling. In accordance with the specific weather conditions, tree stems were collected on 27 July 2003, during typical wet season conditions, and on 18 April 2024, during typical dry season conditions, with one sample per tree. During sampling, nongreen corked twigs with a diameter of 0.3–0.5 cm and a length of 3–5 cm were cut from the sunny side of *Eucalyptus*, quickly peeled, placed in sampling bottles, and sealed with Parafilm. The samples were promptly placed in a portable icebox at approximately 4 °C and subsequently delivered to the lab for freezing and kept at −20 °C.

2.2.3. Collection of Soil Samples

The collection time of soil samples is consistent with that of tree stem samples. During stem sample collection, two soil samples were gathered adjacent to the sampled plants at various depths: 0–20 cm, 20–40 cm, 40–60 cm, 60–90 cm, and 90–150 cm by soil drill. These soil samples were obtained to assess the soil water content and stable hydrogen isotopes in soil water. Among these samples, the soil samples tested for isotopic composition were put in a bottle that was hermetically wrapped with a Parafilm membrane. The preservation of the soil samples was consistent with that of the tree stem samples. Another portion of the soil sample was stored in the aluminium box.

2.3. Sample Processing and Isotope Analysis

The soil sample in the aluminium box was subsequently subjected to drying at 105 °C for 12 h to analyse the soil water content. Tree xylem and soil water were extracted using the evaporative cooling method through a vacuum extraction apparatus (LI-2100, LICA, Beijing, China) [20]. To assure the accuracy of sample analysis and eliminate the influence of organic pollution, a 13 mm diameter organic filter with a minuscule pore size of 0.45 µm was used to filter the extracted water sample. The stable hydrogen isotopes of all water samples, including precipitation, soil water, and xylem water, were examined with a liquid water isotope analyser (LWIA-45-EP, Los Gatos Research, Santa Clara, CA, USA). Hydrogen isotope ratios were expressed in per mil (‰) notation compared with V-SMOW (Vienna Standard Mean Ocean Water), and the estimation accuracy of the LGR (Los Gatos Research) instrument was <0.2‰ for δD [21].

2.4. Data Analysis

Firstly, we used the direct inference method to ascertain the water sources of *Eucalyptus*. In this study, the xylem water of *Eucalyptus* was exclusively derived from soil water at varying depths. The method is applied to determine the primary water-absorbing soil layers of *Eucalyptus* by analysing the isotopic intersection locations between the xylem water and soil water lines [22]. Secondly, the Bayesian mixed model (MixSIAR) (version 3.1.7) was used to determine the proportion of water uptake by *Eucalyptus* from each soil layer. The calculation process included the following steps. The δD values of xylem water were uploaded into the MixSIAR model as mixed data, and the means and standard deviations of δD from the soil water in each layer were input into the MixSIAR model as source data. The δD fractionation coefficient was then set to 0, which means that there was no fractionation of isotopes during water transport from the root to the xylem [23]. The running length of “Markov chain Monte Carlo, MCMC” was set to “long”; furthermore, “Specify prior” and “Error structure” were set to “Uninformative prior” and “Residual only”, respectively. The model was evaluated for convergence via the “Gelman–Rubin” and “Geweke” methods [24]. A one-way ANOVA was conducted to investigate the xylem water δD among *Eucalyptus* of varying ages ($p < 0.05$).

3. Results

3.1. Meteorological and Environmental Conditions

During the entire research period (April 2023 to June 2024), the total precipitation was 1844.2 mm, with the precipitation primarily concentrated in the wet season from May to October, totalling 862.4 mm. It should be noted that between April 2023 and March 2024, the precipitation was 1168 mm, which is close to the perennial mean precipitation (1286 mm) in the study area. In the dry season (November 2023 to April 2024), the precipitation was only 286.6 mm, indicating a notable disparity in precipitation between the dry and wet seasons (Figure 2). Specifically, in the typical wet season month of June 2023, the precipita-

tion reached 231.7 mm, accounting for 26.9% of the total precipitation in the wet season. Conversely, in December 2023, which is in the middle of dry season, the precipitation was 17.7 mm, constituting only 6.2% of the total precipitation in the dry season. Similar to precipitation, the daily average temperature notably varied between the dry and wet seasons. In the wet season, the daily average temperature was relatively high, measuring 26.6 °C and ranging from 19.3 to 32.3 °C. While in the dry season, the daily average temperature was lower, at 17.9 °C, but with a wider range of variation, ranging from 7.0 to 28.7 °C. Therefore, it is evident that the study area experiences greater precipitation and higher temperatures during the wet season, whereas the opposite situation was manifested in the dry season, indicating obvious characteristics of rain and heat in the same season. In addition, the average δD value of rainfall from May to October was -63.20‰ , whereas that from November to April of the following year was -9.70‰ . There was substantial variation in the δD values between the dry and wet seasons, with a positive deviation observed during the dry season and a negative deviation during the wet season.

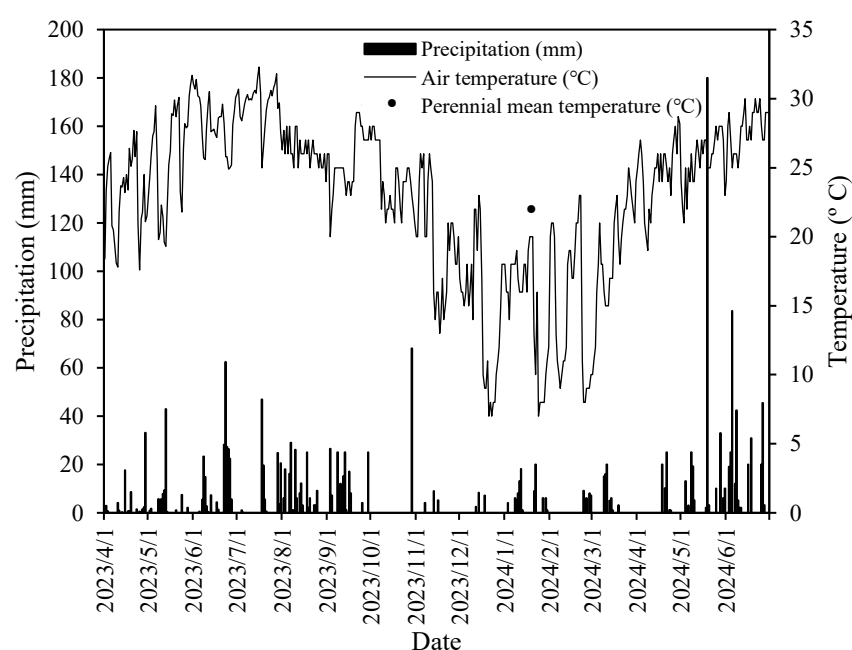


Figure 2. Daily precipitation and average air temperature from April 2023 to June 2024: growing season (April to October) and non-growing season (November to March).

3.2. Variation in Soil Water Content in Stands of Different Ages During Dry and Wet Seasons

This study revealed that the surface soil (0–20 cm) in *Eucalyptus* plantations of different ages was characterized by a low water content in both the dry and wet seasons (Figure 3). However, as the soil layer deepened, the soil water content gradually increased in the wet and dry seasons. In the dry season, the soil water content in April ranged from 14.8% to 26.9%, averaging 22.5% within the 0–150 cm soil layers. The average soil water content in the 0–20 cm surface layer was only 17.0%, indicating a relative deficiency of soil water in the dry season. During the wet season, the soil water content in August ranged from 25.1% to 34.9%, averaging 31.1% within the 0–150 cm soil profile, which was significantly greater than the average soil water content in the dry season ($p < 0.05$). The average soil water content at depths of 0–20 cm was as high as 28.3%, indicating that the soil water was recharged in a timely manner by ample precipitation during the wet season and that the soil water content remained high.

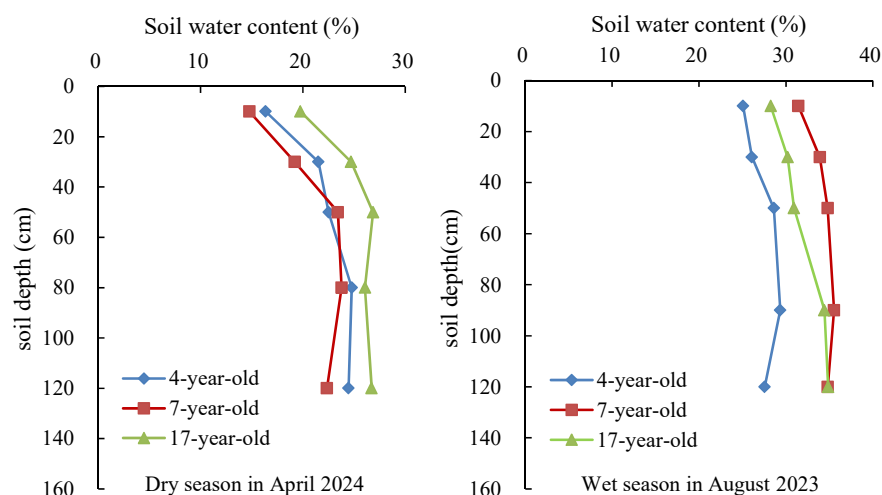


Figure 3. Dynamic changes in the soil water content of *Eucalyptus* plantations of different ages during dry and wet seasons.

3.3. δD in Soil and Xylem Water

Figure 4 showed that the soil water δD values varied between the dry and wet seasons in the *Eucalyptus* plantations of different ages. Specifically, in the dry season, the δD of soil water in the 4-year-old and 7-year-old *Eucalyptus* plantations gradually decreased with increasing soil depth, whereas the soil water δD of the 90–150 cm deep soil layer in the 17-year-old *Eucalyptus* plantations was higher than that of the upper 0–40 cm layer. In the wet season, the soil water δD in the 0–20 cm was slightly greater in the 7-year-old *Eucalyptus* plantation than that in the 90–150 cm deep soil. Conversely, the opposite was observed in the 17-year-old *Eucalyptus* plantation.

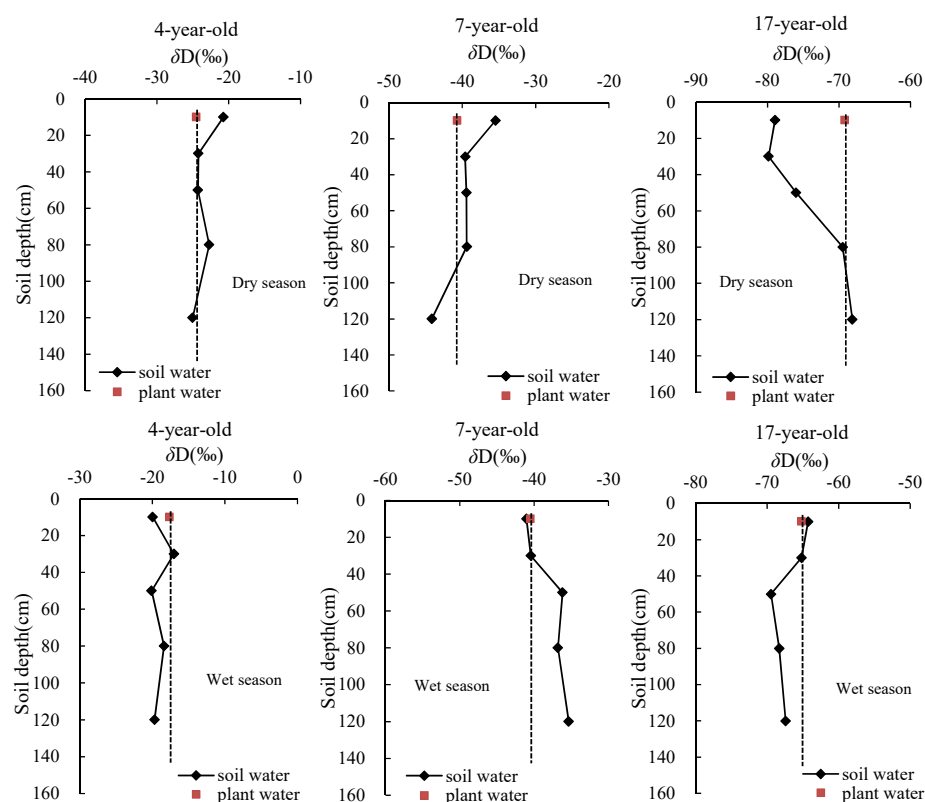


Figure 4. δD in xylem water and soil water in *Eucalyptus* plantations of different ages in August 2023 and April 2024.

The δD of xylem water in *Eucalyptus* plantations of different ages significantly varies during the dry and wet seasons, indicating that there was a difference in the water sources utilized by *Eucalyptus* in different seasons. In April, during the dry season, the xylem water δD of the 4-year-old *Eucalyptus* intersected with the δD of the soil water in the 20–60 cm and 90–150 cm layers, indicating that 4-year-old *Eucalyptus* primarily absorbed soil water from the 20–40 cm, 40–60 cm and 90–150 cm layers in the dry season. In addition, the xylem water δD of the 7-year-old and 17-year-old *Eucalyptus* closely matched the soil water δD in the 90–150 cm soil layer, suggesting that 7-year-old and 17-year-old *Eucalyptus* mainly utilized soil water in the 90–150 cm and 60–90 cm soil layers in the dry season. In August, during the wet season, the xylem water δD of the 4-year-old *Eucalyptus* intersected with the soil water δD of the 20–40 cm and 40–60 cm layers, suggesting that 4-year-old *Eucalyptus* primarily utilized soil water from the 20–60 cm soil layer in the wet season. The xylem water δD from 7-year-old and 17-year-old *Eucalyptus* closely resembled the soil water in the 0–40 cm soil layer, indicating that 7-year-old and 17-year-old *Eucalyptus* mainly relied on soil water from the 0–40 cm soil layer in August.

3.4. Water Use Patterns of *Eucalyptus* of Different Ages

The results of the MixSIAR model indicated significant differences in the utilization of soil water at distinct depths by *Eucalyptus* of different ages during the dry and wet seasons (Figure 5). In April, during the dry season, *Eucalyptus* aged 4, 7, and 17 years primarily absorbed water from the 40–150 cm soil layer. Among them, the highest proportion existed in water uptake from the 60–150 cm soil layer by 17-year-old *Eucalyptus*, with an absorption proportion of 50.9%. However, 4-year-old and 7-year-old *Eucalyptus* mainly absorbed water from the 40–90 cm soil layer, with absorption proportions of 40.6% and 40.8%, respectively. The utilization ratios of surface soil water (0–20 cm) were relatively low in the *Eucalyptus* stands of different ages, with the lowest utilization ratio of 0–20 cm soil water observed in the 17-year-old *Eucalyptus*, at only 15.5%.

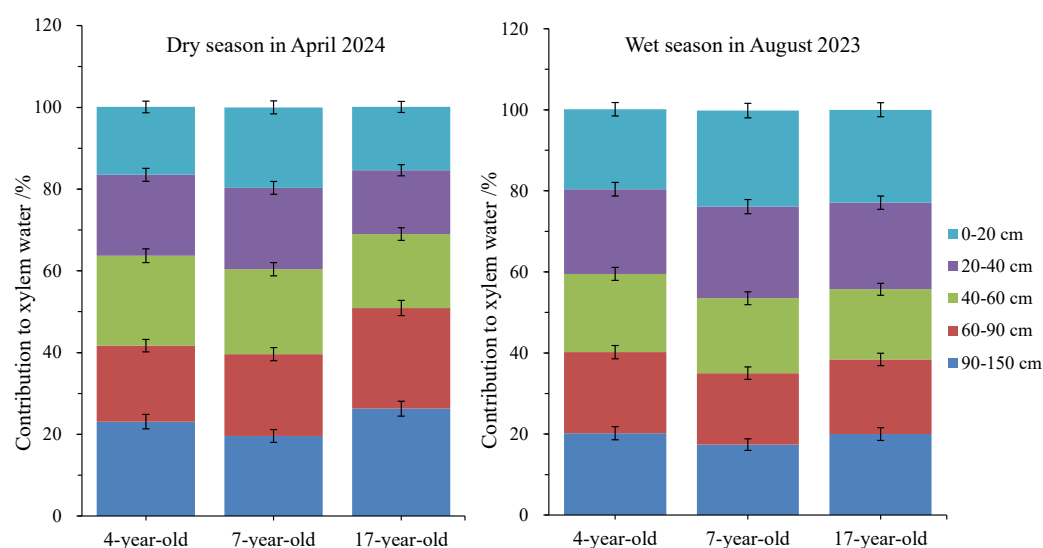


Figure 5. The contributions of each soil layer to the xylem water of *Eucalyptus* of different ages in August 2023 and April 2024.

In August, during the wet season, the water sources for *Eucalyptus* stands of different ages varied from those in the dry season. All the trees relied primarily on soil water at a 0–60 cm depth. Among the stands, the 7-year-old *Eucalyptus* utilized a relatively high proportion of water uptake from the upper 0–20 cm and 20–40 cm soil layers, with values of 23.7% and 22.6%, respectively. For 17-year-old *Eucalyptus*, most of the soil water was

obtained from the 0–20 cm soil layer, at 22.9%, then the 20–40 cm soil layer followed at 21.4%. The water obtained from the 40–60 cm soil layer was the lowest, at 17.3%. For 4-year-old *Eucalyptus*, compared with the dry season, the proportion of soil water obtained from the 0–20 cm soil layer slightly increased in the wet season, at 19.7%, whereas the proportion of deep soil water utilized in the 90–150 cm soil layer decreased.

These findings clearly indicated that in the dry season, *Eucalyptus* plants of different ages exhibited relatively low water utilization from the 0–20 cm soil layer. Instead, they primarily depended on water from the 40–150 cm soil layer. During the wet season, *Eucalyptus* mainly utilized soil water from layers above 60 cm. Notably, 7-year-old *Eucalyptus* obtained 64.8% water from the 0–60 cm soil layer.

4. Discussion

4.1. Soil Water Content and Hydrogen Isotopes of Soil Water in *Eucalyptus* Plantations of Different Ages During Dry and Wet Seasons

Our study found that surface soil has a low water content and a wide range of variation in *Eucalyptus* plantations of different ages, which is consistent with previous research [25]. This phenomenon may be attributed to the severe impact of surface evaporation on the surface soil [26]. As the soil layer deepens, the impact of the external environment gradually diminishes, resulting in an increase in the soil water content and a tendency towards stabilization. In addition, the average soil moisture within the 0–150 cm profile in August considerably exceeded that in April ($p < 0.05$), particularly in the 0–20 cm surface soil. This observation indicated that the soil water deficit in the dry season was more severe than that in the wet season, especially in the surface soil. The main reason may be related to factors including rainfall, soil characteristics, and local meteorological conditions [27–29]. Specifically, the precipitation of the study area in the wet season is significantly higher than that in the dry season; therefore, the soil moisture in the wet season is higher than that in the dry season. Meanwhile, the evaporation of the surface soil layer during the dry season is stronger than that of the other soil layers, resulting in severe water deficit in the surface soil during the dry season.

Our results also showed that the δD values of the soil water in every layer gradually decreased with increasing soil depth in the 4-year-old and 7-year-old *Eucalyptus* plantations in the dry season. This phenomenon is chiefly caused by the significant evaporative enrichment of stable isotopes in surface soil water [30–33]. However, the δD values of the soil water within the 0–150 cm profile of the 17-year-old *Eucalyptus* plantation exhibited the opposite trend in the dry season. This observation suggests that while evaporation weakly enriched the δD values of deep soil water, the presence of older soil water (referred to as old water), the distribution of roots, strong water absorption capacity, and high clay content and water holding capacity contributed to the elevated δD values of soil water [34,35]. During the wet season, the δD values of the soil water showed slight fluctuation in the various layers (0–150 cm) of the 4-year-old *Eucalyptus* plantation. The δD value of soil water increased with increasing soil depth in the 7-year-old *Eucalyptus* plantation, which can be attributed to the mixing and dilution of rainwater isotopes and initial soil water isotopes during the infiltration process [36]. However, in the 17-year-old *Eucalyptus* plantation, the δD value of the soil water decreased following the increase in soil layers. This may be influenced by factors such as lateral groundwater recharge and soil texture.

4.2. Water Use Patterns of *Eucalyptus* of Different Ages During Dry and Wet Seasons

The distribution of precipitation in the subtropical region of southern China during dry and wet seasons is uneven, resulting in fluctuations in water resource availability for plants during different growing seasons [37–39]. Plants generally alter their water

usage strategies to adapt to these changes in water availability, and different plants may exhibit various water use adaptations during dry and wet seasons [40–42]. In this study, *Eucalyptus* plantations of different ages primarily utilized soil water in the 40–150 cm layer during the dry season. However, the main depth of water uptake varied slightly among the different age groups of the *Eucalyptus* plantations. The 4-year-old and 7-year-old *Eucalyptus* plantations mainly relied on soil water in the 40–90 cm soil layer, whereas the 17-year-old *Eucalyptus* plantation primarily absorbed deeper soil water in the 60–150 cm layer. This may be because of the absence of rainfall for an extended period and dry air during the dry season, which, combined with high surface evaporation, led to reductions in soil moisture levels in the surface layer. As a result, the utilization of deep soil water by *Eucalyptus* plantations of different ages increased. Furthermore, given that the root distribution of the 17-year-old *Eucalyptus* was deeper and wider than that of the 4-year-old and 7-year-old trees, it demonstrated a greater ability to utilize soil water in the 60–150 cm layer [43,44]. In contrast, the main water-absorbing soil layer of *Eucalyptus* significantly shifted towards the upper soil layer during the wet season. The primary reason for this is that the frequent rainfall during the wet season effectively replenishes shallow soil water, resulting in a greater availability of shallow soil water. Plants require less energy to absorb water through surface roots; hence, they prioritize absorbing water stored in shallow soil to minimize energy consumption. This pattern of water utilization is similar to that observed in 9-year-old *Eucalyptus* on the Leizhou Peninsula [19]. This similarity may be attributed to comparable climate characteristics, tree species, and soil conditions between tropical and subtropical regions.

The tropical and subtropical regions of South China are highly vulnerable to climate change. In recent years, the frequency of extreme weather events, such as seasonal droughts, has increased in this region, which poses substantial risks to local plantation ecosystems [45,46]. This study aimed to reveal the water utilization patterns of *Eucalyptus* plantations of different ages. The results can provide a scientific foundation for the accurate management of plantation vegetation and the efficient utilization of water resources in South China. Furthermore, they can contribute to the formulation of environmental risk management strategies in anticipation of future climate change.

5. Conclusions

In this study, stable hydrogen isotopes and a MixSIAR model were primarily utilized to examine the water uptake patterns of *Eucalyptus* at different ages during dry and wet seasons. The findings indicated that there was a notable difference in the water use pattern of *Eucalyptus* of different ages between the dry and wet seasons. In the dry season, 4-year-old and 7-year-old *Eucalyptus* primarily utilized soil water in the 40–90 cm soil layer, with utilization ratios of 40.6% and 40.8%, respectively, and 17-year-old *Eucalyptus* mainly absorbed water from the 60–150 cm soil layer, with an absorption ratio of 50.9%, while the remaining 49.1% was predominantly sourced from the 0–60 cm soil layer. During the wet season, 4-year-old, 7-year-old, and 17-year-old *Eucalyptus* primarily utilized the upper soil water at depths of 0–60 cm, with utilization ratios of 59.9%, 64.8%, and 61.6%, respectively. These findings demonstrate that following varying water source availability in the environment, *Eucalyptus* could flexibly alter its water uptake strategy, indicating a strong ability to cope with seasonal drought induced by climate change.

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